AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

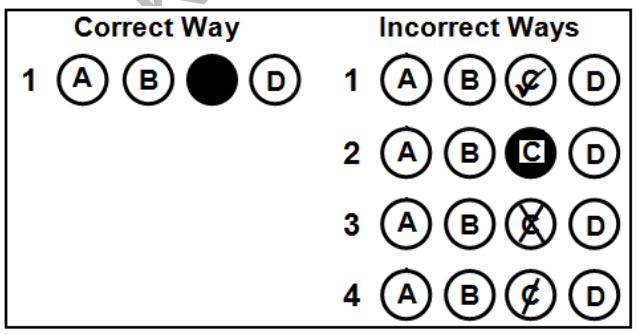
Business Mathematics Paper I

Time: 30 minutes Marks: 20

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 20 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. There are 1.5 dozen bananas, 8 apples, 4 pears and 30 apricots in a fruit basket. The ratio of bananas to apple in simple ratio is

   A. 9:4
   B. 4:9
   C. 8:1.5
   D. 1.5:8

2. A person read a book having 1850 pages in 25 days while reading 3 hours per day. The number of pages he read in 17 days at the same rate will be

   A. 222
   B. 616
   C. 1258
   D. 2720

3. If 34% of a number is 69.36, then this number will be

   A. 45.78
   B. 49.01
   C. 204.00
   D. 235.80

4. If the annual interest rate is 20% and the interest is compounded semi-annually, then the interest per period is

   A. 5%
   B. 10%
   C. 20%
   D. 40%

5. Ahmed invested Rs 4,000 for 2 years and earned a simple interest of Rs 200 on it. The rate of interest is

   A. 2.5%
   B. 5%
   C. 7.5%
   D. 10%

6. Ms Nasreen borrowed a sum of Rs 150,000 from bank at an interest rate of 10% per annum compounded annually. After how many years, the amount payable to the bank will become Rs 181,500?

   A. 2.0 years
   B. 2.5 years
   C. 3.0 years
   D. 3.5 years

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7. $(12)_8 - (6)_8$ is equal to

A. $(3)_8$
B. $(5)_8$
C. $(4)_8$
D. $(6)_8$

8. The binary equivalent of 17 is

A. $(10001)_2$
B. $(10011)_2$
C. $(11001)_2$
D. $(10101)_2$

9. $(444)_5 + (134)_5$ is equal to

A. $(223)_5$
B. $(578)_5$
C. $(1023)_5$
D. $(1133)_5$

10. Which of the following ordered pairs lie(s) on the *y*-axis?

I. (0, 0)
II. (3, 0)
III. (0, 3)

A. I only
B. II only
C. I and III
D. II and III

11. Which of the following is one of the binary relations from $B = \{5, 7, 9\}$ to $A = \{1, 2, 3\}$?

A. {(1, 5), (2, 5), (3, 5)}
B. {(7, 5), (2, 7), (9, 3)}
C. {(5, 5), (5, 7), (5, 9)}
D. {(5, 1), (5, 2), (5, 3)}

12. The equation of the line passing through $(0, -2)$ and having slope -2 is

A. $2x + y - 2 = 0$
B. $x + 2y + 2 = 0$
C. $2x + y + 2 = 0$
D. $x + 2y - 2 = 0$

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13. The value of x, which satisfies the equation, $ax + a = b$ is

A. $\frac{b}{a^2}$

B. $\frac{a-b}{a}$

C. $\frac{b-a}{a}$

D. $b - 2a$

14. The solution set of simultaneous linear equations $\begin{matrix} x + y = c \\ x - y = 0 \end{matrix}$ is

A. $\{(0, c)\}$

B. $\{(c, c)\}$

C. $\left\{\left(\frac{c}{2}, 0\right)\right\}$

D. $\left\{\left(\frac{c}{2}, \frac{c}{2}\right)\right\}$

15. The solution set of a quadratic equation $x^2 + 2x = 0$, is

A. $\{0, 2\}$

B. $\left\{0, \frac{1}{2}\right\}$

C. $\{0, -2\}$

D. $\left\{0, -\frac{1}{2}\right\}$

16. The factors of $ax^2 - bx^2 - by + ay$ are

A. $(a-b)(x^2 - y)$

B. $(a-b)(y - x^2)$

C. $(b+a)(x^2 - y)$

D. $(a-b)(x^2 + y)$

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17. Which of the following matrix multiplications is possible?

A. $\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

B. $\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a \\ b \end{bmatrix}$

C. $\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$

D. $\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a & b \end{bmatrix}$

18. For matrix $A = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, the determinant is

A. -1
B. 0
C. a
D. $-a$

19. A system of linear equations $x - y + 1 = 0$ and $y = 0$ can be written in matrix equation as

A. $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

B. $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

20. The matrix $\begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ is

I. Diagonal matrix
II. Square matrix
III. Scalar matrix

A. I only
B. II only
C. I and II
D. I and III

END OF PAPER

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